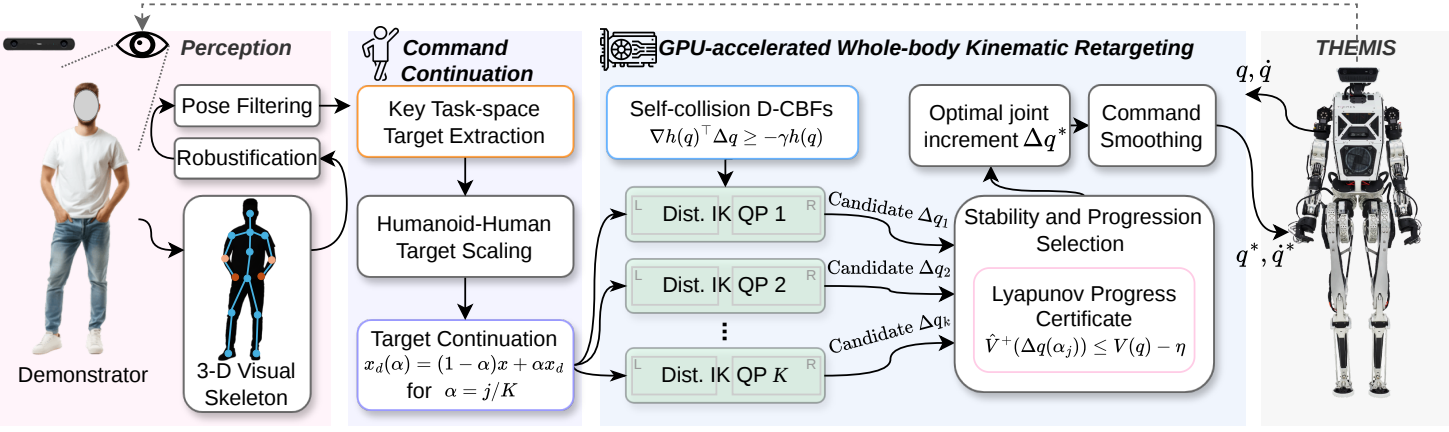
Perception
Pose Filtering
Robustification
Demonstrator
3-D Visual Skeleton
Command Continuation
Key Task-space Target Extraction
Humanoid-Human Target Scaling
Target Continuation
$x_d(\alpha) = (1-\alpha)x + \alpha x_d$
for $\alpha = j/K$
GPU-accelerated Whole-body Kinematic Retargeting
Self-collision D-CBFs
$\nabla h(q)^\top \Delta q \geq -\gamma h(q)$
Dist. IK QP 1
Dist. IK QP 2
Dist. IK QP K
Candidate Δq_1
Candidate Δq_2
Candidate Δq_k
Stability and Progression Selection
Lyapunov Progress Certificate
$\hat{V}^+(\Delta q(\alpha_j)) \leq V(q) - \eta$
Optimal joint increment Δq^*
Command Smoothing
THEMIS
$q, \dot{q}$
$q^*, \dot{q}^*$